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AE498 HW2

Code Explanation:

This block of code initializes all the parameters and creates Keplerian element arrays.

```python
# Constants
AU = 149597870 #km
MU_SUN = 1.32712440041279419e11 # km3/s2
MU_EARTH = 398600 # km3/s2

EARTH_Q = 9.78969676e-01 * AU # km
EARTH_E = 1.83889630e-02
EARTH_A = EARTH_Q / (1 - EARTH_E)
EARTH_I = 7.55697209e-05 # rad
EARTH_ARGP = 5.34322623 # rad
EARTH_RAAN = 2.63906227 # rad
EARTH_M = 1.88972281 # rad
EARTH_PER = np.sqrt(4 * np.pi**2 * EARTH_A**3 / MU_SUN)
EARTH_N = 2 * np.pi / EARTH_PER

FI2026_Q = 0.74161536 * AU # km
FI2026_E = 0.19293483
FI2026_A = FI2026_Q / (1 - FI2026_E)
FI2026_I = 0.05877868 # rad
FI2026_ARGP = 2.22509624 # rad
FI2026_RAAN = 3.55784498 # rad
FI2026_M = 3.64624167 # rad
FI2026_PER = np.sqrt(4 * np.pi**2 * FI2026_A**3 / MU_SUN)
FI2026_N = 2 * np.pi / FI2026_PER

EARTH_KEP = np.array([EARTH_A, EARTH_E, EARTH_I, EARTH_ARGP, EARTH_RAAN, EARTH_M])
FI2026_KEP = np.array([FI2026_A, FI2026_E, FI2026_I, FI2026_ARGP, FI2026_RAAN, FI2026_M])
```

✓ 0.0s

These blocks of code were taken from work in AE402. They convert Keplerian elements to vecs and vecs to elements.

```python
#Newton's method applied to find eccentric anomaly
def find_E(M, e):
    E_prev = M
    E_curr = E_prev - (E_prev - e * np.sin(E_prev) - M) / (1 - e * np.cos(E_prev))
    while (abs(E_curr - E_prev) > 1E-10):
        E_prev = E_curr
        E_curr = E_prev - (E_prev - e * np.sin(E_prev) - M) / (1 - e * np.cos(E_prev))
    return E_curr

#Elements to vectors calculator
def elems_to_vecs(kep_state, mu):
    """
    Converts classical orbital elements to position and velocity vectors.
    Inputs:
        a   : Semi-major axis (float)
        e   : Eccentricity (float)
        i   : Inclination (float, radians)
        ran : Right ascension of ascending node (float, radians)
        arg : Argument of periapsis (float, radians)
        M   : Mean anomaly (float, radians)
        mu  : Gravitational parameter of the central body (float)
    Outputs:
        r : Position vector (numpy array)
        v : Velocity vector (numpy array)
    """

    a, e, i, arg, ran, m = kep_state

    #Find eccentric anomaly and convert to true
    E = find_E(m, e)
    f = 2*np.arctan2(np.sqrt(1+e)*np.sin(E/2), np.sqrt(1-e)*np.cos(E/2))
    theta = arg + f

    #Solve for angular momentum and r_mag
    h = (mu * a * (1 - e**2))**(1/2)
    r_mag = (a * (1 - e**2)) / (1 + e * np.cos(f))

    #I, J, K vecs
    I = np.array([1, 0, 0])
    J = np.array([0, 1, 0])
    K = np.array([0, 0, 1])

    #Solve for r
    ehat_r = ((np.cos(theta) * np.cos(ran) - np.cos(i) * np.sin(ran) * np.sin(theta)) * I +
              (np.cos(theta) *np.sin(ran) + np.cos(i) * np.cos(ran) * np.sin(theta)) * J +
              (np.sin(i) * np.sin(theta)) * K)
    r = r_mag * ehat_r

    #Solve for v
    v = (-mu / h * (np.cos(ran) * (np.sin(theta) + e * np.sin(arg)) + np.sin(ran) * (np.cos(theta) + e * np.cos(arg)) * np.cos(i)) * I +
         -mu / h * (np.sin(ran) * (np.sin(theta) + e * np.sin(arg)) - np.cos(ran) * (np.cos(theta) + e * np.cos(arg)) * np.cos(i)) * J +
         mu / h * (np.cos(theta) + e * np.cos(arg)) * np.sin(i) * K)
    v_mag = np.sqrt(v.dot(v))

    return np.array([r[0], r[1], r[2], v[0], v[1], v[2]])
```

```python
# Old AE402 Function I am reusing
def vecs_to_elems(r, v, mu):
    """
    Converts position and velocity vectors to classical orbital elements.
    Inputs:
        r : Position vector (numpy array)
        v : Velocity vector (numpy array)
        mu: Gravitational parameter of the central body (float)
    Outputs:
        a   : Semi-major axis (float)
        e   : Eccentricity (float)
        i   : Inclination (float, radians)
        ran : Right ascension of ascending node (float, radians)
        arg : Argument of periapsis (float, radians)
        f   : True anomaly (float, radians)
    """
    #Solve for various values to be used
    r_mag = np.linalg.norm(r)
    v_mag =np.linalg.norm(v)
    h = np.cross(r, v)
    h_mag = np.linalg.norm(h)
    I = np.array([1, 0, 0])
    J = np.array([0, 1, 0])
    K = np.array([0, 0, 1])
    n = np.cross(K, h)
    n_mag = np.linalg.norm(n)

    #Get a, e, i
    a = 1 / (2 / r_mag - v_mag**2 / mu)
    e = (v_mag**2 / mu - 1 / r_mag) * r - 1 / mu * (r.dot(v)) * v
    e_mag = np.linalg.norm(e)
    i = np.acos((h / h_mag).dot(K))

    #Get ran, arg, f
    ran = np.acos(n.dot(I) / n_mag)
    if n.dot(J) < 0:
        ran = 2 * np.pi - ran
    arg = np.acos(n.dot(e) / (n_mag * e_mag))
    if e.dot(K) < 0:
        arg = 2 * np.pi - arg
    f = np.acos(r.dot(e) / (r_mag * e_mag))
    if r.dot(v) < 0:
        f = 2 * np.pi - f

    E = 2 * np.arctan2(np.sqrt(1 - e_mag) * np.sin(f / 2), np.sqrt(1 + e_mag) * np.cos(f / 2))
    M = E - e_mag * np.sin(E)

    return np.array([a, e_mag, i, arg, ran, M])
```

This block of code implements gauss's planetary equations as an ODE which can be integrated with RK4.

```python
def gauss_state_deriv(kep_state, a_dr, a_dt, a_dh, dt):
    """
    Returns the step of keplerian states
    Inputs:
        kep_state : a, e, i, argp, raan, m (km, unitless, rad, rad, rad, rad)
        a_dr : perturbing radial acceleration (km/s2)
        a_dt : perturbing tangential acceleration (km/s2)
        a_dh : perturbing normal acceleration (km/s2)
        dt : time step (s)
    Outputs:
        kep_state_deriv
    """
    a, e, i, argp, raan, m = kep_state

    # Various needed values
    E = find_E(m, e)
    f = np.arctan2(np.sqrt(1 - e**2) * np.sin(E), np.cos(E) - e)
    theta = argp + f
    p = a * (1 - e**2)
    r = p / (1 + e * np.cos(f))
    h = np.sqrt(MU_SUN * p)
    b = np.sqrt(p / a)
    per = np.sqrt(4 * np.pi**2 * a**3 / MU_SUN)
    n = 2 * np.pi / per

    d_raan = r * np.sin(theta) / (h * np.sin(i)) * a_dh
    d_i = r * np.cos(theta) / h * a_dh
    d_argp = 1 / (h * e) * (-p * np.cos(f) * a_dr + (p + r) * np.sin(f) * a_dt) - r * np.sin(theta) * np.cos(i) / (h * np.sin(i)) * a_dh
    d_a = 2 * a**2 / h * (e * np.sin(f) * a_dr + p / r * a_dt)
    d_e = 1 / h * (p * np.sin(f) * a_dr + (p + r) * np.cos(f) * a_dt)
    d_m = n + b / (a * h * e) * ((p * np.cos(f) - 2 * r * e) * a_dr - (p + r) * np.sin(f) * a_dt)

    return np.array([d_a, d_e, d_i, d_argp, d_raan, d_m])
```

This block of code integrates the Keplerian elements of Earth and FI2026 forwards using my own implementation of RK4. Commented out is a transformation of Earth's perturbation to the radial, transverse, and out of plane frame. This was added as perturbative accelerations to FI2026. In class, however, it was discussed that this level of detail was not necessary.

```python
def gauss_rk4_step(earth_state, body_state, a_arr, rv_earth, rv_body, dt):
    """
    Complets a step of gaussian planetary propagation in rk4
    Inputs:
        earth_state : keplerian params a, e, i, argp, raan, m
        body_state : keplerian params a, e, i, argp, raan, m
        rv_earth : position and velocity of earth
        rv_body : position and velocity of body
        a_arr : np array of accelerations [a_dr, a_dh, a_dt]
    """

    k1 = gauss_state_deriv(earth_state, 0, 0, 0, dt)
    k2 = gauss_state_deriv(earth_state + 0.5 * dt * k1, 0, 0, 0, dt)
    k3 = gauss_state_deriv(earth_state + 0.5 * dt * k2, 0, 0, 0, dt)
    k4 = gauss_state_deriv(earth_state + dt * k3, 0, 0, 0, dt)
    next_earth = earth_state + (dt / 6.0) * (k1 + 2*k2 + 2*k3 + k4)

    a_dr, a_dh, a_dt = a_arr[0], a_arr[1], a_arr[2]
    # --------- Second body perturbation calculation (no longer need per class instructions) ------
    # r_body = rv_body[0:3]
    # v_body = rv_body[3:6]
    # r_body_to_earth = rv_earth[0:3] - rv_body[0:3]
    # rad = r_body / np.linalg.norm(r_body)
    # out = np.cross(r_body, v_body) / np.linalg.norm(np.cross(r_body, v_body))
    # tan = np.cross(out, rad) / np.linalg.norm(np.cross(out, rad))
    # a_second_body = MU_EARTH * r_body_to_earth / np.linalg.norm(r_body_to_earth)**3

    # a_dr += np.dot(a_second_body, rad)
    # a_dt += np.dot(a_second_body, tan)
    # a_dh += np.dot(a_second_body, out)

    k1 = gauss_state_deriv(body_state, a_dr, a_dt, a_dh, dt)
    k2 = gauss_state_deriv(body_state + 0.5 * dt * k1, a_dr, a_dt, a_dh, dt)
    k3 = gauss_state_deriv(body_state + 0.5 * dt * k2, a_dr, a_dt, a_dh, dt)
    k4 = gauss_state_deriv(body_state + dt * k3, a_dr, a_dt, a_dh, dt)
    next_body = body_state + (dt / 6.0) * (k1 + 2*k2 + 2*k3 + k4)

    return next_earth, next_body
```

This block of code does the full trajectory propagation for Earth and FI2026 and returns an array of times, Keplerian states, and position-velocity states.

```python
def gauss_rk4_propagate(earth_state0, body_state0, a_arr, t0, tf, dt):
    n_steps = int(np.ceil((tf - t0) / dt))
    t = t0

    earth_state = earth_state0.copy()
    rv_earth = elems_to_vecs(earth_state, MU_SUN)

    body_state = body_state0.copy()
    rv_body = elems_to_vecs(body_state, MU_SUN)

    time_list  = [t]
    earth_state_list = [earth_state.copy()]
    body_state_list = [body_state.copy()]
    earth_rv_list = [rv_earth]
    body_rv_list = [rv_body]
    second_body_list = []

    # I asked ChatGPT to help me make a progress bar, that is why this looks a bit weird
    for _ in tqdm(range(n_steps), desc="Propagating"):

        # Adjust final step to land exactly on tf
        step = min(dt, tf - t)
        if step <= 0:
            break

        earth_state, body_state = gauss_rk4_step(earth_state, body_state, a_arr, rv_earth, rv_body, step)
        rv_earth = elems_to_vecs(earth_state, MU_SUN)
        rv_body = elems_to_vecs(body_state, MU_SUN)

        t += step

        time_list.append(t)

        earth_state_list.append(earth_state.copy())
        body_state_list.append(body_state)
        earth_rv_list.append(rv_earth)
        body_rv_list.append(rv_body)

    return np.array(time_list), np.array(earth_state_list), np.array(body_state_list), np.array(earth_rv_list), np.array(body_rv_list)
```

This code does b-plane plotting given earth's velocity, FI2026's relative position to earth, and FI2026's relative velocity all at the time of collision. First, the xi and zeta coordinates are calculated. Then, the Valsecchi circles, if ratios are given, are solved for in plotted. The methods used here are less complicated than those in Valsecchi's paper as relative velocity is already known and does not need to be calculated.

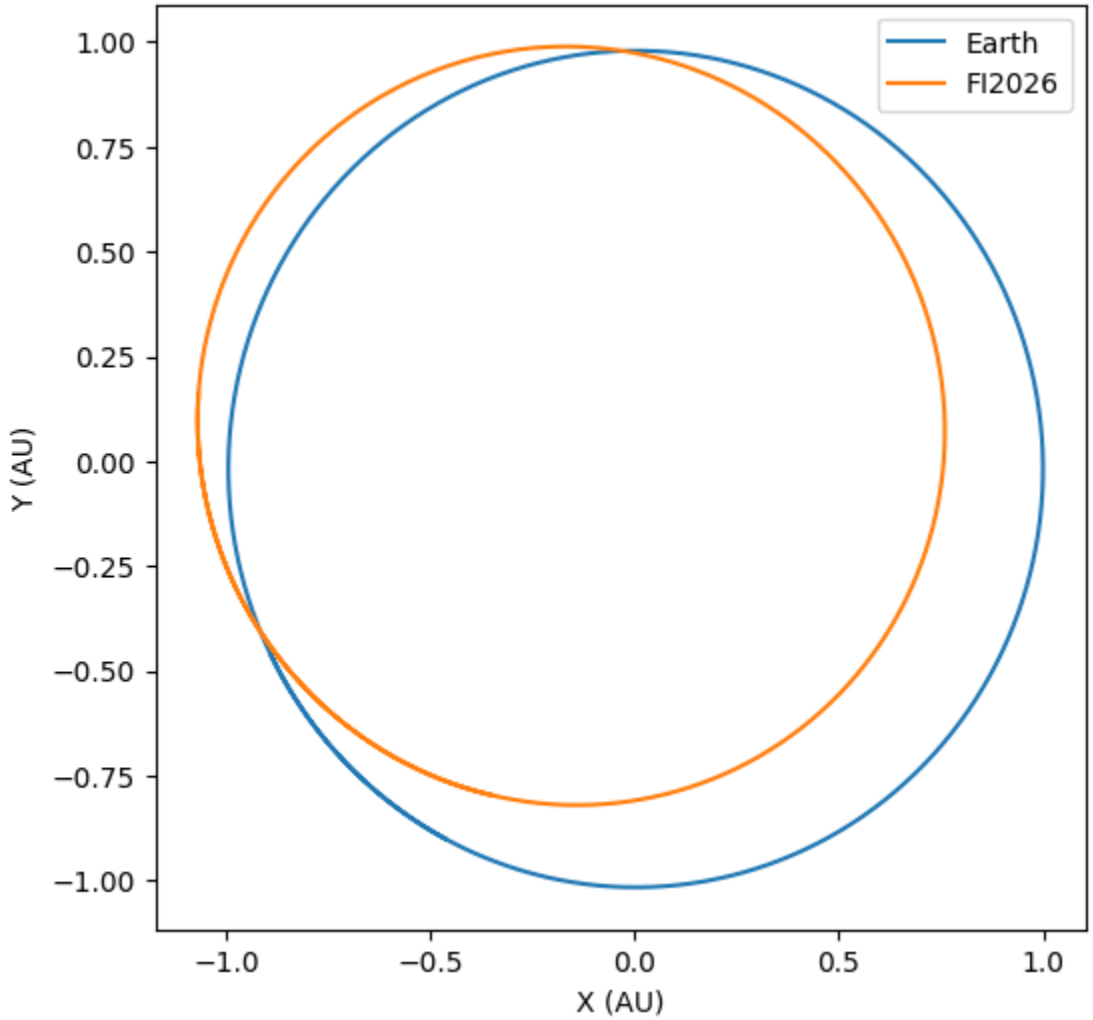
```
from matplotlib.patches import Circle

def plot_bplane(earth_vel, rel_pos, rel_vel, title = "", valsecchi_ratios = []):
    """
    Plot b plane with impact location and gravitational focusing of Earth
    Inputs
        earth_vel - 1d np array
        rel_vel - 1d np array
        rel_pos - 1d np array
        body kep - state array a, e, i, argp, raan, i
        title - title of the graph
        valsecchi_rations - resonance ratios T_body / T_planet
    """
    R_E = 6378 # km
    MU_E = 398600 # km3/s2
    s_hat = rel_vel / np.linalg.norm(rel_vel)
    xi_hat = np.cross(earth_vel, s_hat) / np.linalg.norm(np.cross(earth_vel, s_hat))
    zeta_hat = np.cross(-s_hat, xi_hat)
    h = np.cross(rel_pos, rel_vel)
    b = np.cross(s_hat, h) / np.linalg.norm(rel_vel)
    xi = np.dot(b, xi_hat)
    zeta = np.dot(b, zeta_hat)
    b_max = R_E * np.sqrt(1 + 2 * MU_E / (R_E * np.linalg.norm(rel_vel)**2))

    fig, ax = plt.subplots(figsize = (8, 6))

    U_kms = np.linalg.norm(rel_vel)
    c = MU_E / U_kms**2
    theta = np.acos(np.dot(rel_vel, earth_vel) / np.linalg.norm(rel_vel) / np.linalg.norm(earth_vel))

    color_list = ['red', 'green', 'brown', 'indigo', 'violet']

    for i, vr in enumerate(valsecchi_ratios):

        U = np.linalg.norm(rel_vel) / np.linalg.norm(earth_vel)
        ap_body = (vr)**(2 / 3)
        theta_p = np.acos((1 - U**2 - 1 / ap_body) / (2 * U))

        D = (c * np.sin(theta)) / (np.cos(theta_p) - np.cos(theta))
        R = abs((c * np.sin(theta_p)) / (np.cos(theta_p) - np.cos(theta)))

        vc = Circle((0, D / R_E), R / R_E, label = f'VC {vr:.3f}', color = color_list[i], fill = False)
        ax.add_patch(vc)


    earth = Circle((0, 0), R_E / R_E, color = 'blue', fill = False, label = "Earth")
    earth_focused = Circle((0, 0), b_max / R_E, color = 'grey', fill = False, label = "Earth Focused")
    ax.add_patch(earth)
    ax.add_patch(earth_focused)
    ax.scatter(zeta / R_E, xi / R_E, s = 30, color = 'orange', label = "FI2026")
    ax.set_xlabel(r"$\zeta$ (Earth Radii)")
    ax.set_ylabel(r"$\xi$ (Earth Radii)")
    ax.set_title(title)
    ax.legend()
    ax.plot()
```

This block of code is an example of how perturbations were added to FI2026. You can see how there is a separate a_arr and dv_arr for continuous and instantaneous accelerations. The instantaneous velocity change is converted to x-y-z coordinates before being added to FI2026's velocity. Once FI2026's new position and velocity are calculated, they are transformed back to Keplerian elements. The continuous acceleration is simply fed into the Gauss planetary equations. The collision index is found as the time where relative position is lowest. This works because Earth is not a perturbing acceleration, so FI2026 is on the b-plane when distance is minimized. Admittedly, large time steps may skip over the exact b-plane time, but 1000s was found to work and have minimal changes on the b-plane plot.

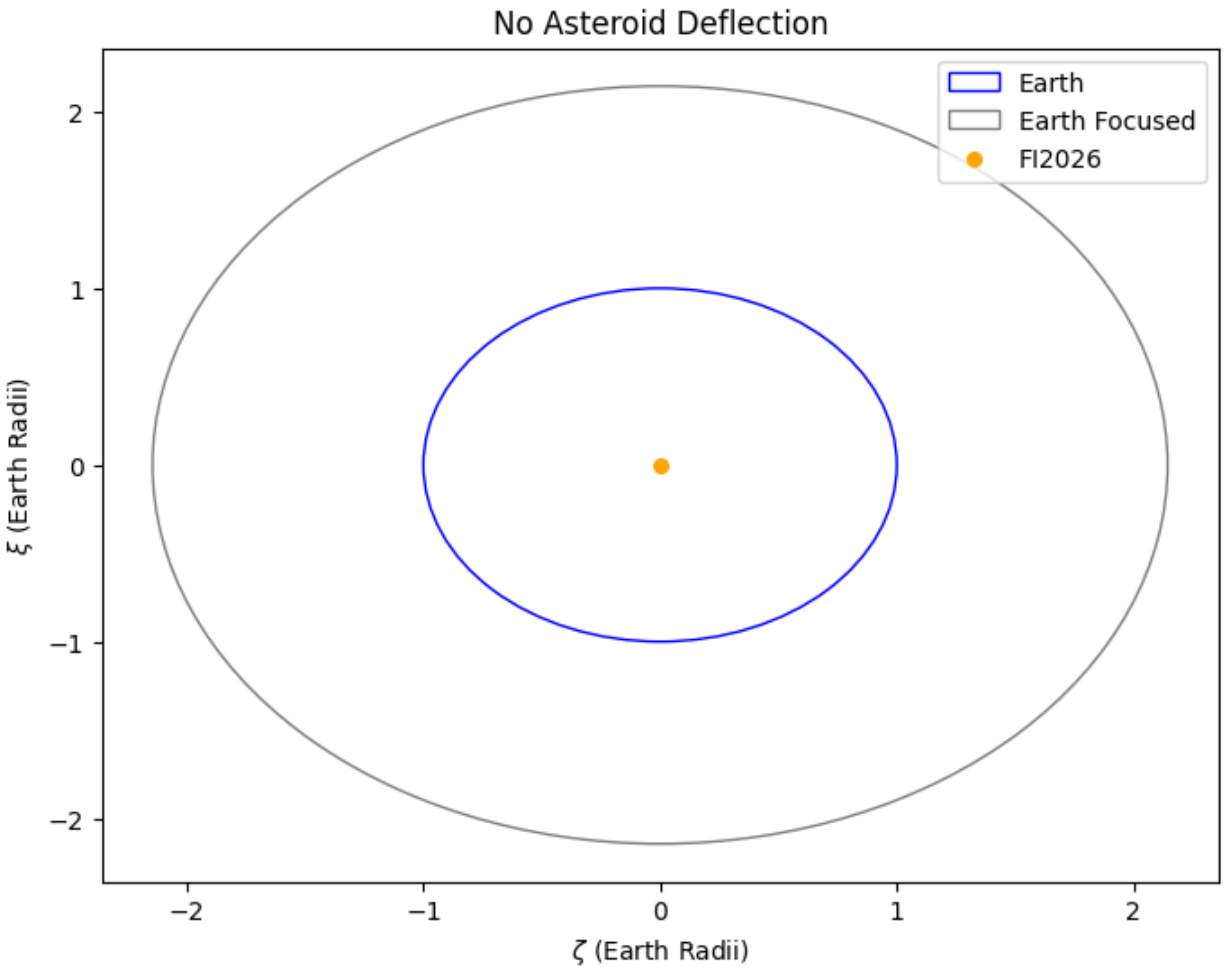
```
t0 = 0
tf = 3600 * 24 * 400
dt = 1000

a_arr = np.array([0, 0, 0])
dv_arr = np.array([0.01, 0, 0])
FI2026_STATE = elems_to_vecs(FI2026_KEP, MU_SUN)

# solve for radial, out of plane, and tangential, dv_dr, dv_dh, dv_dt
r_body = FI2026_STATE[0:3]
v_body = FI2026_STATE[3:6]
rad = r_body / np.linalg.norm(r_body)
out = np.cross(r_body, v_body) / np.linalg.norm(np.cross(r_body, v_body))
tan = np.cross(out, rad) / np.linalg.norm(np.cross(out, rad))
dv_arr_xyz = dv_arr[0] * rad + dv_arr[1] * out + dv_arr[2] * tan

FI2026_STATE[3:6] = FI2026_STATE[3:6] + dv_arr_xyz

FI2026_KEP_IMPACT = vecs_to_elems(FI2026_STATE[0:3], FI2026_STATE[3:6], MU_SUN)
TIME_LIST, EARTH_STATE_LIST, FI2026_STATE_LIST, EARTH_RV_LIST, FI2026_RV_LIST = gauss_rk4_propagate(EARTH_KEP, FI2026_KEP_IMPACT, a_arr, t0, tf, dt)

# Find collision index
R_DIFF_LIST = EARTH_RV_LIST - FI2026_RV_LIST
R_DIFF_MAG_LIST = np.linalg.norm(R_DIFF_LIST, axis = 1)
collision_idx = np.argmin(R_DIFF_MAG_LIST)

collision_earth_pos = EARTH_RV_LIST[collision_idx][0:3]
collision_fi2026_pos = FI2026_RV_LIST[collision_idx][0:3]
collision_rel_pos = collision_fi2026_pos - collision_earth_pos

collision_earth_vel = EARTH_RV_LIST[collision_idx][3:6]
collision_fi2026_vel = FI2026_RV_LIST[collision_idx][3:6]
collision_rel_vel = collision_fi2026_vel - collision_earth_vel
```

Question 1:

The time to collision = 360 days

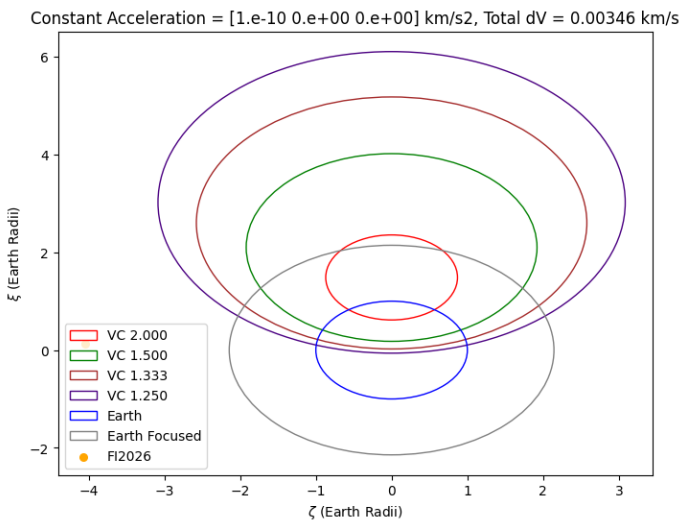


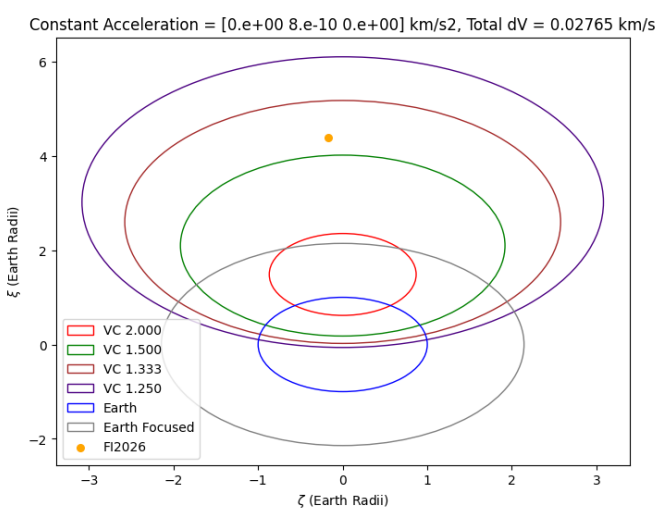
Question 2:

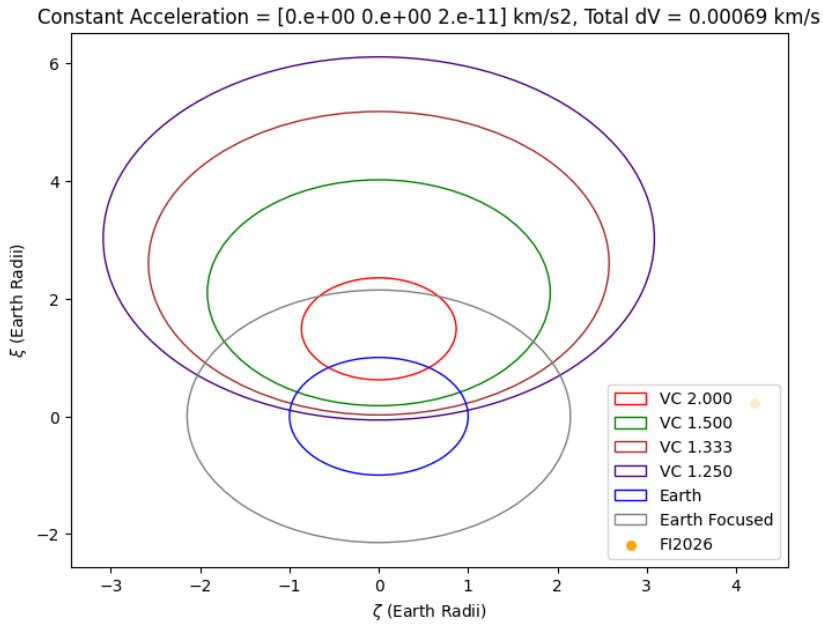

No Asteroid Deflection
Earth
Earth Focused
FI2026
2
1
0
−1
−2
ξ (Earth Radii)
−2
−1
0
1
2
ζ (Earth Radii)

Question 3:

To model continuous perturbations, Gauss's planetary equations are used. I iterated on each of the three axes in gauss's planetary equations (radial, out of plane, and transverse) until the asteroid was 4 Earth radii away. The FI2026 marker is hard to find in the first and third image, it is behind the legend. It is easiest to deflect the asteroid when pushing it transversally, needing a continuous acceleration of just 2e-8 m/s2. The equivalent delta-v over this time period was just 0.69 m/s. For an asteroid weighing 7.3e10 kg like Bennu, this is just a 365N continuously applied.

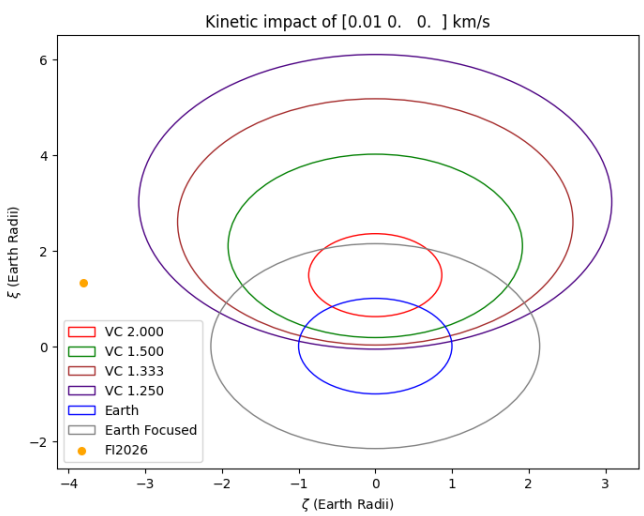


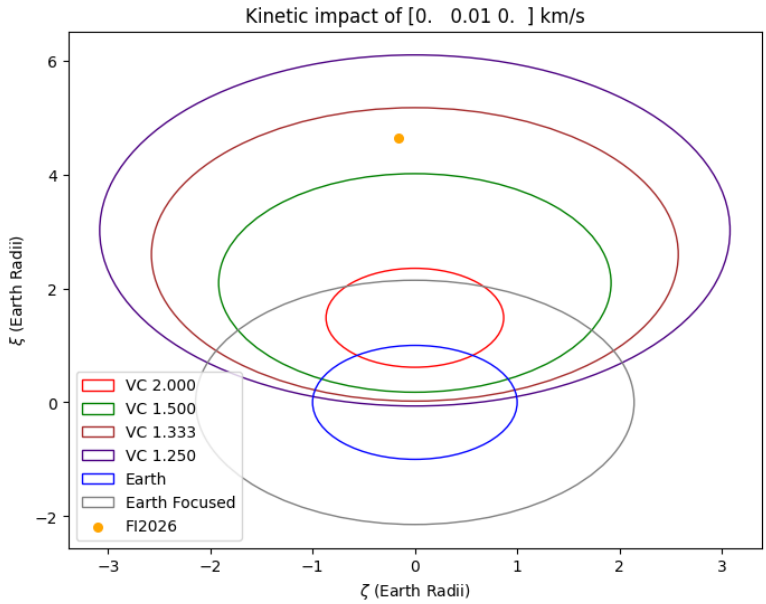


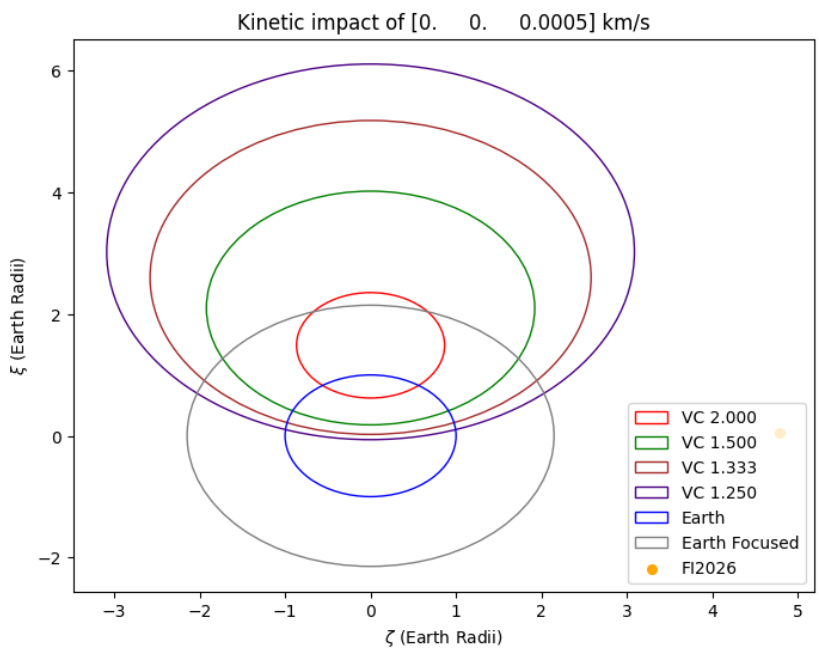


Question 4:

To model a kinetic impact, the velocity of FI2026 is simply adjusted before propagation begins. Velocity was added in the same frame as the earlier continuous acceleration (radial, out of plane, and transverse) and iterated until a distance of 4 Earth radii was achieved. Guass's planetary equations were used again for propagation, but perturbative accelerations were set to zero. Again, some of the asteroid impact locations are behind the legend. Overall, less total delta-v is needed for kinetic impacts as frontloading velocity changes increases the amount of time it contributes to position changes.







Question 5:

Propagations were done for each direction in the previous two questions. The easiest direction for perturbation is transverse, roughly aligned with the asteroid's velocity. This makes sense as changing the angular momentum through out of plane acceleration is naturally difficult. For an asteroid the size of Bennu, a Delta-V of 0.5 m/s requires 3.6e10 N of instantaneous force.

Question 6:

Valsecchi circles for 2:1, 3:2, 4:3, 5:4 ratios are plotted in each of the plots above. Only the out of plane velocity changes get close to hitting any of the Valsecchi circles. Radial and transverse velocity move the b-plane intersection to the right or left.